

New determination of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ S-factor

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We present new measurements of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ cross section from $\bar{E}_{\text{cm}} = 116$ to 2460 keV. Our new measurements lead to $S_{17}(0) = 22.1 \pm 0.6(\text{expt}) \pm 0.6(\text{theor}) \text{ eV b}$ based on data from $\bar{E}_{\text{cm}} = 116$ to 362 keV, where the central value is based on the theory of Descouvemont and Baye. We compare our results to other $S_{17}(0)$ values extracted from both direct (${}^7\text{Be}(p,\gamma){}^8\text{B}$) and indirect (Coulomb dissociation and heavy-ion reaction) measurements, and show that the results of these 3 types of experiments are not mutually compatible. We recommend a “best” value, $S_{17}(0) = 21.4 \pm 0.5(\text{expt}) \pm 0.6(\text{theor}) \text{ eV b}$, based on the mean of all modern direct measurements below the 1^+ resonance.

1. INTRODUCTION

Precise predictions of the *production rate* of ${}^8\text{B}$ solar neutrinos are important for limiting the allowed neutrino mixing parameters including possible contributions of sterile species and also for testing solar models. The predicted ${}^8\text{B}$ production rate in the sun is based on solar model calculations that depend on measured reaction rates for the solar burning steps following the initial p+p fusion, the most uncertain of which is the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ rate. The currently recommended value for $S_{17}(0)$, the astrophysical S-factor for this reaction, is $19^{+4}_{-2} \text{ eV b}$ [1]. This value needs to be known to better than $\pm 5\%$ in order that its uncertainty not be important in the predicted ν_e production rate [2].

2. THE SEATTLE-TRIUMF DIRECT MEASUREMENTS

We have measured the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ cross section over a range of 3 orders of magnitude in three separate experiments. The ${}^7\text{Be}$ targets were produced at TRIUMF, while the cross section measurements were conducted at the University of Washington Center for Experimental Nuclear Physics and Astrophysics. We eliminated a major uncertainty in

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many previous experiments due to uncertain and nonuniform target areal density by using a beam magnetically-rastered to $7 \times 7 \text{ mm}^2$, in order to produce a nearly uniform flux over a small $\sim 3.5 \text{ mm}$ diameter target. A cryopumped closed vacuum system with multiple LN_2 cold traps provided a clean vacuum. We directly measured and monitored the energy-loss profile of the target using a narrow ${}^7\text{Be}(\alpha, \gamma){}^{11}\text{C}$ resonance. That also showed that carbon built up on the target surface was negligible. We made frequent *in situ* measurements of the ${}^7\text{Be}$ target activity to determine the target sputtering losses. We also made the first (and, to date, the only) measurements of ${}^8\text{B}$ backscattering losses.

The first experiment (BE1) [3] used a target of 106 mCi initial activity to measure the ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$ cross section at mean center-of-mass proton energies, $\bar{E}_{\text{c.m.}} = 186$ to 1203 keV, including the M1 resonance near $\bar{E}_{\text{c.m.}} = 630 \text{ keV}$. The α -detector solid angle in this experiment was determined using a ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ reaction yield ratio. In a second, abbreviated measurement with a new ${}^7\text{Be}$ target of similar activity, we determined the cross section in the range $\bar{E}_{\text{c.m.}} = 876$ to 2459 keV. In this second measurement (BE2), we became concerned over possible inaccuracies in the ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ reaction yield ratio method for determining the α -detector solid angle, which relied on calculated extrapolations of the continuous α -spectrum tail below the experimental cutoff (a similar but smaller extrapolation is necessary to interpret the ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$ reaction yields). Recently we completed the third measurement (BE3) with a 340 mCi target which covered the range $\bar{E}_{\text{c.m.}} = 116$ to 1754 keV, for details, see ref. [4].

The BE3 experiment included several improvements on previous measurements and yielded our best determination of the absolute cross section. Hence we based our absolute S-factor determination on this experiment. We avoided using the ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ reaction for solid-angle normalization, using instead a custom made ${}^{148}\text{Gd}$ α -source deposited on a backing of the same design as the one used for the ${}^7\text{Be}$ target. We made measurements both with a large area detector (L) (area $\approx 416 \text{ mm}^2$, thickness $\approx 33 \mu\text{m}$) and with a smaller, thinner Si surface-barrier detector (S) (area $\approx 139 \text{ mm}^2$, thickness $\approx 20 \mu\text{m}$) for which the correction and hence the uncertainty due to the fraction of the ${}^8\text{B}$ β -delayed α spectrum lost below the detector threshold was minimal.

The target was mounted on a rotating water cooled arm. The arm was rotated by a computer controlled servo motor, and the counting cycle timing was determined by a hardwired electronic sequencer box. Following irradiation, the target was rotated to a shielded location where the α 's from the ${}^8\text{B}$ decays were detected. During the α -counting period the proton beam flux through a 3 mm aperture was measured. A single (p, γ) cross-section measurement corresponded to a large number of such cycles. Details of the cross-section determination and S-factor calculation are given in ref. [4].

Figure 1 shows the results of our experiments in the energy range below $\bar{E}_{\text{c.m.}} = 1800 \text{ keV}$. A fit with the Descouvemont and Baye [5] model plus a 1^+ resonance describes the data very well over a large range in energy, with small deviations near $\bar{E}_{\text{c.m.}} \approx 1200 \text{ keV}$. The low energy data provide the most reliable basis for determining $S_{17}(0)$, since the 1^+ resonance is avoided and the model dependence (and hence also the uncertainty in the extrapolation to zero energy) is minimized. The experimental uncertainty due to the α -threshold correction is also minimized, and the high energy tail of the 1^+ resonance is avoided.

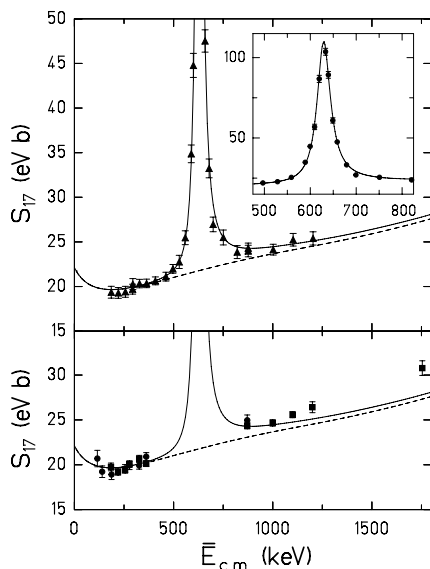


Figure 1. Top panel: BE1 data renormalized to $S_{17} = 22.1$ eV b. The inset shows the resonance region. The solid curves show the best-fit Descouvemont & Baye curve plus a fitted 1^+ resonance. The dashed curve shows the DB model only. Bottom panel: solid squares BE3 small detector data, solid circles BE3 large detector data. The solid curve was calculated with 1^+ resonance parameters fixed from the fit to the BE1 data, and the normalization was determined by fitting the BE3 data with $\bar{E}_{c.m.} \leq 362$ keV.

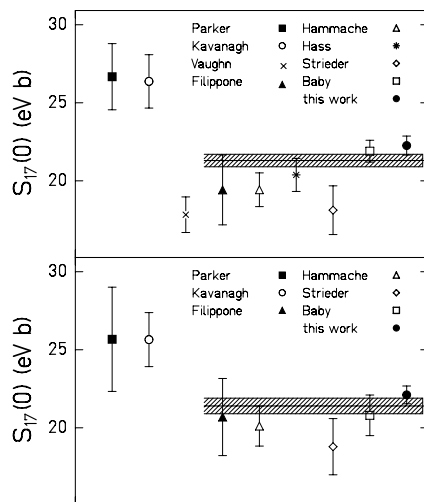


Figure 2. $S_{17}(0)$ values determined from our DB fits to published data from direct experiments. Bottom panel (top panel): fits to data with $\bar{E}_{c.m.} \leq 425$ keV (1200 keV). The horizontal line and shaded bands indicate the mean values and uncertainties determined from fitting the data of Filippone and more recent experiments.

The weighted average of the small and large detector data below $\bar{E}_{c.m.} = 362$ keV yield

$$S_{17}(0) = 22.1 \pm 0.6(\text{expt}) \pm 0.6(\text{theor}) \text{ eV b} , \quad (1)$$

including a common-mode scale factor uncertainty of $\pm 2.3\%$, and an extrapolation uncertainty determined as the RMS deviation of fits to the BE3 data with 12 different theories. This result is in excellent agreement with our previously published value [3] and supercedes it. Details of the uncertainty determination can be found in ref. [4].

3. COMPARISON WITH OTHER DIRECT MEASUREMENTS

We compare the results of all direct experiments by fitting the DB theory to published data in two different energy ranges: $\bar{E}_{c.m.} \leq 425$ keV, and $\bar{E}_{c.m.} \leq 1200$ keV. We made a substantial effort to ensure accuracy in these comparisons by obtaining data from primary sources whenever possible, and by fitting the data ourselves. For details of the fits and

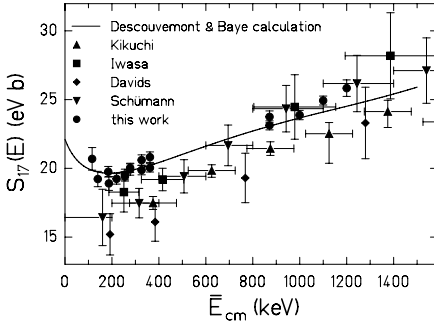


Figure 3. S -factors inferred from Coulomb dissociation experiments compared with our direct results (with the 1^+ resonance subtracted) and the best fit DB curve to our data. The experimental error bars shown are relative and do not include scale-factor uncertainties.

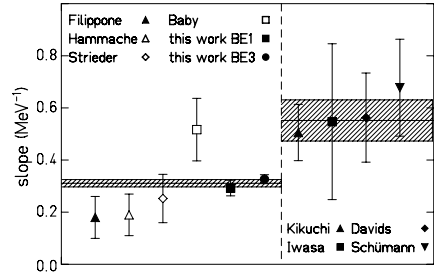


Figure 4. $S_{17}(\bar{E}_{c.m.})$ slopes determined from straight-line fits to direct $S_{17}(\bar{E}_{c.m.})$ data (corrected for the 1^+ resonance tail) (left panel) and to $S_{17}(\bar{E}_{c.m.})$ values inferred from CD experiments (right panel). The horizontal lines and shaded regions correspond to the mean values and uncertainties determined from the direct data and from the CD data, respectively.

references to the other experiments, please see ref. [4]. The fits to low energy data provide the most reliable determination of $S_{17}(0)$. It is also instructive to compare the experiments over a wider range, as some experiments do not have a good precision at low energies. The results for $\bar{E}_{c.m.} \leq 425$ keV are, $S_{17}(0) = 21.4 \pm 0.5$ eV b, $\chi^2/\nu = 1.2$ ($\nu = 4$), and for $\bar{E}_{c.m.} \leq 1200$ keV, $S_{17}(0) = 21.2 \pm 0.4$ eV b, $\chi^2/\nu = 2.4$ ($\nu = 5$). We note the excellent agreement between the results of Baby *et al.* and this work, over both fit ranges. The good agreement between the $S_{17}(0)$ values determined from the 2 different fit ranges demonstrates that the energy dependence of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ cross section is well-determined and similar for the direct experiments, and that the $S_{17}(0)$ value determined here is insensitive to the fit range.

4. COMPARISON WITH COULOMB DISSOCIATION MEASUREMENTS

S_{17} values have been inferred from the Coulomb breakup of ${}^8\text{B}$ (CD) and peripheral heavy-ion transfer and breakup reactions. The ${}^7\text{Be}(p,\gamma){}^8\text{B}$ E1 S_{17} -factors inferred from the 4 most recent CD experiments are shown as a function of $\bar{E}_{c.m.}$ in Fig. 3 compared to our present direct results (with the 1^+ resonance subtracted) and the best-fit DB curve to our data. The CD $S_{17}(E)$ values are similar to our direct values in the region 800 - 1200 keV, while they systematically fall below our values at low energies. Our direct measurements below 1200 keV have an energy dependence that is well-described by the DB theory (plus a 1^+ resonance) except at the highest energies (1100 - 1200 keV) where DB falls a few percent low. The energy dependence inferred from the CD experiments is

significantly steeper than the DB theory and does not agree with the direct results.

We have quantified the difference in energy dependence between direct and CD experiments by fitting a straight line of the form $S_{17}(\bar{E}_{\text{c.m.}}) = a(1 + b\bar{E}_{\text{c.m.}})$ to data in the range $\bar{E}_{\text{c.m.}} \leq 425$ keV and 830 keV $\leq \bar{E}_{\text{c.m.}} \leq 1300$ keV. For the direct data, the M1 component was subtracted to be able to compare the slope to the pure E1 CD S-factors. All of the fits had $\chi^2/\nu < 1.3$, indicating that the straight line is a good approximation. Fig. 4, displays b , the fitted slope and demonstrates a systematic difference between the 2 types of experiments. From the direct experiments the mean slope is 3.11 ± 0.14 MeV⁻¹, with $\chi^2/\nu = 1.9$, and the mean slope determined from the CD data is 5.5 ± 0.8 MeV⁻¹, with $\chi^2/\nu = 0.2$. After increasing the uncertainty on the direct mean by the factor $(1.9)^{1/2}$ to account for the fit χ^2/ν , we find the probability that these 2 results arise from the same parent distribution is $P(\chi^2, \nu) = 0.003$.

5. CONCLUSIONS

The Seattle-TRIUMF experiment measured the cross section of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ reaction from $\bar{E}_{\text{c.m.}} = 116$ to 2460 keV [4]. We determined $S_{17}(0) = 22.1 \pm 0.6(\text{expt}) \pm 0.6(\text{theor})$ eV b based on data from $\bar{E}_{\text{c.m.}} = 116$ to 362 keV and the model of Descouvemont and Baye [5]. The modern direct experiments (Filippone, Hammache, Strieder, Baby, and this work) give very consistent results. For $\bar{E}_{\text{c.m.}} \leq 425$ keV, the combined fit yields $S_{17}(0) = 21.4 \pm 0.5$ (expt) ± 0.6 (theor) eV b, with $\chi^2/\nu = 1.2$ ($P = 30\%$), which is our new recommended value for $S_{17}(0)$. The recent Coulomb dissociation experiments infer a steeper energy dependence for the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ cross section below 1200 - 1500 keV than is observed in the direct experiments. Since the energy dependence of the present direct measurements is unambiguous (at the level of a few percent), this indicates a systematic error in the interpretation of the CD experiments. Our new recommended value of $S_{17}(0)$ has a combined experimental plus theoretical uncertainty of ± 0.8 eV b, or $\pm 4\%$. This represents a considerable improvement over the 1998 recommendation by Adelberger et al. [1] of $S_{17}(0) = 19 \pm 4$ eV b (1σ). If incorporated into the solar model calculation [2], the uncertainty in $S_{17}(0)$ would no longer make an important contribution to the overall uncertainty in the calculated solar neutrino production rate from ${}^8\text{B}$ decay, and the overall uncertainty would be reduced from $\pm 18\%$ to $\pm 14\%$.

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